

Polyphony J. S. Bruna

PHD CANDIDATE · COGNITIVE AND INFORMATION SCIENCES

University of California, Merced

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Education

Research Interests: Neural Networks, Complex & Dynamical Systems, Collective Intelligence, Joint Action, Alignment

University of California, Merced

Merced, CA

PHD COGNITIVE AND INFORMATION SCIENCES

Expected 2027

- Committee: Christopher T. Kello (chair), Michael J. Spivey, and Jeff Yoshimi
- Master's thesis (*en route*): "Synergetic Mechanisms Underlying the Emergence of Alignment in Human Dyadic Interaction"

University of California, Los Angeles

Los Angeles, CA

GRADUATE FELLOW, INSTITUTE FOR PURE AND APPLIED MATHEMATICS

Fall 2024

- Program: Mathematics of Intelligences

Vassar College

Poughkeepsie, NY

BA COGNITIVE SCIENCE AND PHILOSOPHY (Minor: Japanese)

2018 - 2022

- Overall GPA: 3.99/4.00
- Honors: *Phi Beta Kappa*, *Sigma Xi*, *Psi Chi*, *cum laude generali et cum laude in materia subiecta*
- Thesis: "Leveraging Surprise in a Recurrent Neural Network to (de)Construct Morphological Complexity in Japanese" (Advisors: Jan Andrews and Josh de Leeuw)

Professional Experience

MEMBERSHIPS

- 2022-cont. **Phi Beta Kappa**, Academic Honor Society
- 2022-cont. **Sigma Xi**, The Scientific Research Honor Society
- 2021-cont. **Psi Chi**, International Honor Society in Psychology

POSITIONS

- 2024, 2026 **Humane Studies Fellow**, Institute for Humane Studies, George Mason University
- 2022-2023 **Adolph Sutro Fellow**, Vassar College
- 2021-2022 **Intern**, Cognitive Science, Vassar College
- 2021 **Senior Researcher**, Undergraduate Research Summer Institute, Vassar College
- 2019, 2020 **Research Fellow**, Undergraduate Research Summer Institute, Vassar College
- 2018-2021 **Research Assistant**, Cognitive Science, Vassar College

Publications

* Denotes co-principal authorship

IN PREPARATION

Bruna, P. & Gyllingberg, L. (2025). *Cognition without neurons: modeling anticipation in a basal reservoir computer*.
<https://arxiv.org/abs/2505.02114>

PUBLISHED

Kello, C., **Bruna, P.**, & Thao, K. (2026). Contextual assembly of lexical functions in large language models. *Behavior Research Methods* 58(19). <https://doi.org/10.3758/s13428-025-02898-7>

Bruna, P. and Kello, C. (2025). Least Effort and Alignment in Task-Oriented Communication. *Cognitive Science*, 49, Article e70062. <https://doi.org/10.1111/cogs.70062>

de Leeuw, J. R., Andrews, J., Barney, L., Bigler, M., **Bruna, P.**, Chen, Y., Cherry, R., Dowie, D. R., Forbes, E., Haffey, B., Hu, X., Jaklitsch, M., Leopold, N., Lewis, C., MacDonald, D., McShaffrey, C., Nakayama, K., Olstad, W., Peng, R., ... Zhang, L. (2021). Words May Jump-Start Meaning More Than Vision: A Non-Replication of Early ERP Effects in Boutonnet and Lupyan (2015). *Collabra: Psychology*, 7(1). <https://doi.org/10.1525/collabra.29763>

PROCEEDINGS

Bruna, P., Dale, R., Spivey, M., & Kello, C. (2026). Collective decision-making in coupled echo state networks. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 48 (In press).

Nosrati, S., **Bruna, P.**, Spivey, M., & Marghetis, T. (2026). Spatial navigation at the extremes: The collective accomplishment of high-speed, high-risk navigation in Rally Racing. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 48 (In press).

Kello, C. & **Bruna, P.** (2024). Emergent Mental Lexicon Functions in ChatGPT. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 46. <https://escholarship.org/uc/item/5m9098b5>

*Rane, S., ***Bruna, P.**, Sucholutsky, I., Kello, C., & Griffiths, T. (2023). Concept alignment. *1st NeurIPS Workshop on AI meets Moral Philosophy and Moral Psychology (MP2)*. <https://openreview.net/pdf?id=D3GCdfJL4J>

MISCELLANEOUS

Yoshimi, J., Beckmann, P., Udell, D., **Bruna, P.**, & Meyer, T. (2025). Transformer Architectures and LLMs. In Yoshimi, J., Udell, D., Tosi, Z., Hotton, S., Beckmann, P., Cain, E., Gordon, C., Noelle, D. C., **Bruna, P.**, & Meyer, T. (Eds.), *Neural Networks in Cognitive Science*. <https://downloads.jeffyoshimi.net/NeuralNetworksCogsci.pdf>

Abbas, M., Alharthi, D., Balbas, J., Beckelhymer, D., Beylkin, G., **Bruna, P.**, Cartmill, E., Chormai, P., Corea, F., Dai, X., Das, P., Dipoppa, M., Eberle, O., El Hady, A., Evans, J., Feghhi, E., Flack, J., Flores, C., Foster, J., ... Zhu, Y. (2025). White Paper: "Mathematics of Intelligences." *Institute for Pure and Applied Mathematics*. <https://www.ipam.ucla.edu/reports>

Bruna, P. (2019). A Review of Bryan W. Van Norden, Taking Back Philosophy. *Vassar College Journal of Philosophy*, 6. <https://www.vassar.edu/sites/default/files/2021-08/philosophy-journal-spring-2019.pdf>

Presentations

INVITED TALKS

Spr. 2026. *Collective Decision-Making in Coupled Echo State Networks*. Modeling the Social Mind(s): Theory-Based Computational Models (Symposium). The Society for Personality and Social Psychology Annual Convention, Chicago, IL.

Spr. 2026. *Collective Decision-Making in Coupled Echo State Networks*. Social Interaction Lab, Dept. of Linguistics, Stanford University, CA.

Fall 2025. *Collective Decision-Making in Coupled Echo State Networks*. December Graduate Conference. Institute for Humane Studies, George Mason University, Fairfax, VA.

Fall 2024. *Least Effort and Alignment in Task-Oriented Communication*. Dept. of Cognitive Science, Vassar College, Poughkeepsie, NY.

CONTRIBUTED TALKS

Bruna, P. 2026. *Collective Decision-Making in Coupled Echo State Networks*. 48th Annual Conference of the Cognitive Science Society, Rio de Janeiro, Brazil.

Bruna, P. 2026. *Collective Decision-Making in Coupled Echo State Networks*. Mathematics of Intelligences Reunion Conference at Lake Arrowhead, Institute for Pure and Applied Mathematics, University of California, Los Angeles.

Bruna, P. 2025. *Collective Decision-Making in Coupled Echo State Networks*. Cognitive & Information Sciences Brownbag Series, University of California, Merced.

Bruna, P. 2025. *Collective Information Processing in Loosely Coupled Echo State Networks*. Graduate Workshop in Computational Social Science, Santa Fe Institute, New Mexico.

Bruna, P. 2025. *Basal Reservoir Computing*. Philosophy and Cognitive Science Working Group, University of California, Merced.

- Bruna, P.** & Gyllingberg, L. 2024. *Non-neural Pattern Completion in an Adaptive Reservoir Computer*. Mathematics of Intelligences Culminating Conference at Lake Arrowhead, Institute for Pure and Applied Mathematics, University of California, Los Angeles.
- Bruna, P.** 2024. *Least Effort and Alignment in Task-Oriented Communication*. Mathematics of Intelligences Seminar Series, Institute for Pure and Applied Mathematics, University of California, Los Angeles.
- Bruna, P.** 2024. *Concept Alignment*. Mathematics of Intelligences Working Group on Architecture and Evaluation of Generative AI, Institute for Pure and Applied Mathematics, University of California, Los Angeles.
- Bruna, P.** 2024. *Falands-style Reservoir Networks*. Mathematics of Intelligences Working Group on Ecological and Homeostatic Intelligence, Institute for Pure and Applied Mathematics, University of California, Los Angeles.
- Bruna, P.** 2024. *Concept Alignment*. Philosophy and Cognitive Science Working Group, University of California, Merced.
- Bruna, P.** 2023. *Semantics and Syntax Co-emerge in Adaptive Reservoir Network Dynamics*. Diverse Intelligences Summit, University of St. Andrews, Scotland.
- Bruna, P.** 2020. *Hearing That It Is Silent: How to Hallucinate a Non-Perception*. 10th Annual Mid-Hudson Valley Undergraduate Philosophy Conference, Marist College, Poughkeepsie, NY. (Canceled due to COVID-19)

CONTRIBUTED POSTERS

* Presenting author(s); + Mentored undergraduate

- *Nosrati, S., **Bruna, P.**, Spivey, M., & Marghetis, T. (2026). *Spatial navigation at the extremes: The collective accomplishment of high-speed, high-risk navigation in Rally Racing*. 48th Annual Conference of the Cognitive Science Society, Rio de Janeiro, Brazil.
- ***Bruna, P.**, Dale, R., Spivey, M., & Kello, C. 2025. *Coupled Echo State Networks as a Model of Task-Oriented Alignment*. UC Merced Cognitive Science Showcase, Merced, CA.
- ***Bruna, P.**, Dale, R., Spivey, M., & Kello, C. 2025. *Coupled Echo State Networks as a Model of Task-Oriented Alignment*. 47th Annual Conference of the Cognitive Science Society, San Francisco, CA.
- *Gyllingberg, L., Tian, Y., Sumpter, D., & **Bruna, P.** 2025. *Oscillations and Adaptation: Mathematical Models of Non-Neural Cognition*. Biological Systems that Learn, National Institute for Theory and Mathematics in Biology, Chicago, IL.
- ***Bruna, P.** & Kello, C. 2024. *Alignment Through Dyadic Least Effort in Task-Oriented Conversations*. 65th Annual Meeting of the Psychonomic Society, New York City, NY.
- ***Bruna, P.** & Kello, C. 2024. *Alignment Through Dyadic Least Effort in Task-Oriented Conversations*. Modeling Multi-Scale Collective Intelligences, Institute for Pure and Applied Mathematics, University of California, Los Angeles.
- *Kello, C. & ***Bruna, P.** 2024. *Emergent Mental Lexicon Functions in ChatGPT*. 46th Annual Conference of the Cognitive Science Society, Rotterdam, Netherlands.
- **Thao, K., **Bruna, P.**, & Kello, C. 2024. *Exploring Emergent Embodied Intelligence in Large Language Models: Insights from ChatGPT-4*. 15th Annual California Cognitive Science Conference, Berkeley, CA.
- *Rane, S. & ***Bruna, P.** 2023. *Concept alignment*. 1st NeurIPS Workshop on AI meets Moral Philosophy and Moral Psychology (MP2), New Orleans, LA.
- **Leshock, E., **Pena, Y., **Bruna, P.**, & Andrews, J. 2021. *Learned Categorical Perception with Artificially Generated Faces*. 36th Annual URSI Symposium, Vassar College, Poughkeepsie, NY.
- *Bigler, M. & ***Bruna, P.** 2020. *Probing the Methodology and Interpretation of Learned Categorical Perception Research*. 35th Annual URSI Symposium, Vassar College, Poughkeepsie, NY.
- ***Bruna, P.** & *Marks, P. 2019. *Psychophysical Adaptive Procedure: Developing a New, Generalizable Method*. 34th Annual URSI Symposium, Vassar College, Poughkeepsie, NY.

Honors

- 2025 **Best Poster Award - 2nd place**, *Cognitive Science Showcase, University of California, Merced*
Graduate Student Highlight ("Rising Star: How Life's Journey Shaped a Cognitive Scientist's Path to Discovery"),
<https://graduatedivision.ucmerced.edu/Polyphony-J-Bruna>

Awards, Fellowships, & Grants

* Awarded but declined

2026	Residency Funding Award, Institute for Humane Studies	\$ 4,500
	Professional Development Award, University of California, Merced	\$ 1,725
	Graduate Student Travel Award, University of California, Merced	\$ 1,000
	Summer Research Fellowship, University of California, Merced	\$ 4,050
	Humane Studies Fellowship, Institute for Humane Studies	\$ 1,000
2025	Summer Research Fellowship, University of California, Merced	\$ 5,400
	Hayek Fund Travel Award, Institute for Humane Studies	\$ 3,475
	Graduate Workshop in Computational Social Science Scholarship, Santa Fe Institute	\$ 1,000
	Hayek Fund Travel Award, Institute for Humane Studies	\$ 2,400
2024	Humane Studies Fellowship, Institute for Humane Studies	\$ 5,000
	Fall Research Fellowship, University of California, Merced	\$ 6,000
	MacKenzie Scott Graduate Student Travel Award, University of California, Merced	\$ 1,000
	Travel Award, University of California, Merced	\$ 1,800
	Summer Research Fellowship, University of California, Merced	\$ 5,475
2023	Alumni Research Award, Diverse Intelligences Summer Institute	\$ 750
	Fall Research Fellowship, University of California, Merced	\$ 22,500
	Travel Award, University of California, Merced	\$ 800
	Summer Research Fellowship, University of California, Merced	\$ 1,600
2022	Chancellor's Recruitment Fellowship, University of California, Merced	\$ 45,000
	*Foreign Language and Area Studies (FLAS) Fellowship, U.S. Department of Education	\$ 22,500
	Adolph Sutro Fellowship, Vassar College	\$ 10,000
	The Jamie Nisse Greenberg Philosophy Prize, Vassar College	\$ 900
	Critical Language Scholarship for advanced Japanese, alternate, U.S. Department of State; American Councils for International Education	
2021	Frances Lehman Loeb 1928 Scholarship, Vassar College	\$ 44,000
	Summer Research Fellowship, Vassar College	\$ 4,000
2020	Frances Lehman Loeb 1928 Scholarship, Vassar College	\$ 44,000
	Undergraduate Research Summer Institute Fellowship, Vassar College	\$ 5,000
2019	Frances Lehman Loeb 1928 Scholarship, Vassar College	\$ 42,500
	Undergraduate Research Summer Institute Fellowship, Vassar College	\$ 5,000
2018	1956 Memorial Scholarship, Vassar College	\$ 42,000

Teaching Experience

Spr. 2026	Complex Adaptive Systems, Teaching Assistant, Cognitive Science, University of California, Merced
Fall 2025	Mind, Brain, and Computation, Teaching Assistant, Cognitive Science, University of California, Merced
Spr. 2025	Critical Reasoning, Teaching Assistant, Philosophy, University of California, Merced
Spr. 2024	Contemporary Moral Problems, Teaching Assistant, Philosophy, University of California, Merced
Spr. 2022	Introduction to Cognitive Science, Teaching Assistant, Cognitive Science, Vassar College
Fall 2021	Introduction to Cognitive Science, Teaching Assistant, Cognitive Science, Vassar College

Outreach & Professional Development

WORKSHOPS & TRAINING

- 2026 **Santa Fe Institute**, Advanced Graduate Workshop in Computational Social Science, *Santa Fe, New Mexico*
Diverse Intelligences Summer Institute, *Geneva, NY*
- 2025 **Santa Fe Institute**, Graduate Workshop in Computational Social Science, *Santa Fe, New Mexico*
Complexity Science Hub, Winter School on Complex Systems in Our Changing World, *Vienna, Austria*
- 2024 **Institute for Pure and Applied Mathematics**, Modeling Multi-Scale Collective Intelligences, *University of California, Los Angeles*
Berggruen Institute, Future Humans Workshop on Computational Consciousness, *Beverly Hills, California*
Institute for Pure and Applied Mathematics, Naturalistic Approaches to Artificial Intelligence, *University of California, Los Angeles*
Institute for Humane Studies, Aristotle and Human-Centered AI, *Webinar*
Institute for Pure and Applied Mathematics, Theory and Practice of Deep Learning, *University of California, Los Angeles*
Dynamic Interactions and Methodologies Symposium, *University of California, Los Angeles*
Institute for Pure and Applied Mathematics, Analyzing High-dimensional Traces of Intelligent Behavior, *University of California, Los Angeles*
- 2023 **Santa Fe Institute**, Complexity-GAINs International Summer School on Intelligence and Representation, *Isaac Newton Institute for Mathematical Sciences, University of Cambridge*
Diverse Intelligences Summer Institute, *University of St. Andrews*
- 2022 **Phi Beta Kappa**, Key Connections, *Webinar*
- 2020 **Center for Philosophy of Science**, Pittsburgh Summer Program in Philosophy of Science and Specialized Sciences, *University of Pittsburgh and Carnegie Mellon University*

SERVICE POSITIONS

- 2025-cont. **Book Club President**, Cognitive and Information Sciences, *University of California, Merced*
- 2025 **Brownbag Organizer**, Cognitive and Information Sciences, *University of California, Merced*
- 2023-2024 **Graduate Student Chair**, Cognitive and Information Sciences, *University of California, Merced*
Research Mentor, Learning Aligned Employment Program, *University of California, Merced*
- 2021 **Research Mentor**, Undergraduate Research Summer Institute, *Vassar College*
- 2018-2019 **Editor**, Vassar College Journal of Philosophy, *Vassar College*

PEER REVIEW

Cognition

Cognitive Science

Journal of Experimental Psychology: General

Minds and Machines

SKILLS

Programming:	Python, R, JavaScript, HPC parallel computing
Open source software:	crqa (R package, contributor)
Languages:	English (native), Japanese (basic)